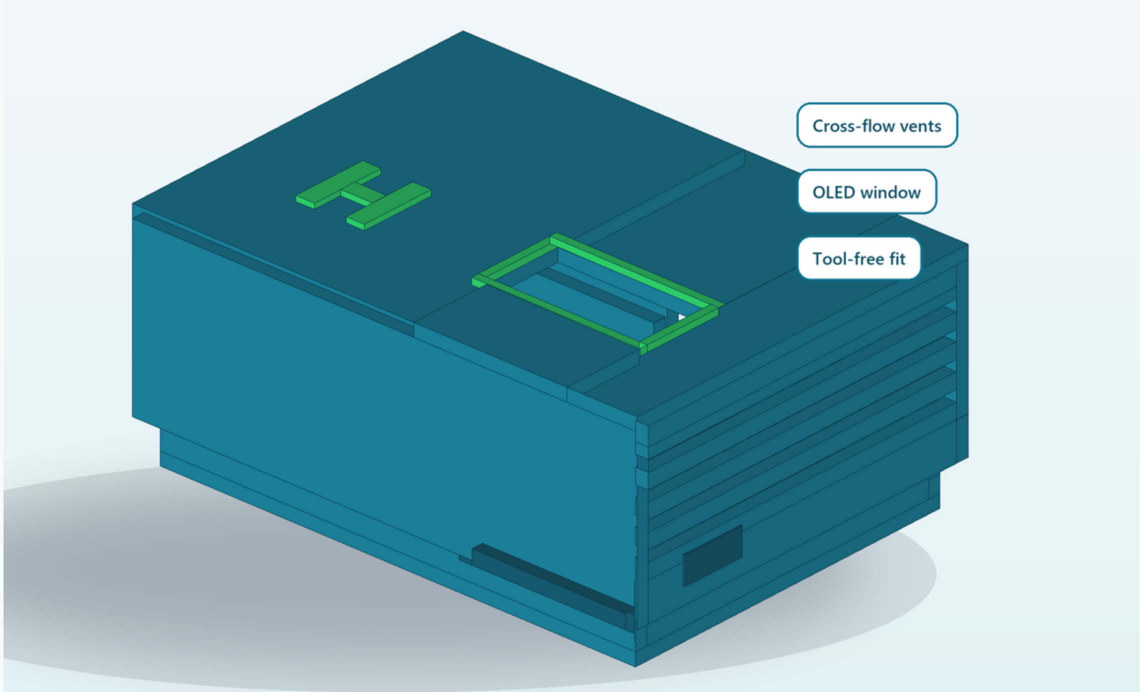
HAWAGUARD MINI

PROJECT OVERVIEW

Original, AI-assisted 3D-printable environmental sensing enclosure

AI-ASSISTED CAD COMPLETE | FILES READY | PHYSICAL PROTOTYPE PENDING



THE IDEA

Affordable environmental sensing becomes more useful when the device is easy to assemble, protect, and maintain. HawaGuard Mini is an original, AI-assisted two-part enclosure concept prepared for a classroom air-quality monitor using an ESP32-class controller, a compact particulate sensor, a temperature/humidity sensor, and a small OLED display. Cross-flow vents support air sampling, while the removable lid makes classroom maintenance simple.

PROJECT WORK COMPLETED

- Defined the use case, size constraints, airflow, and serviceability requirements.
- Created the two-part parametric geometry and internal mounting bays.
- Exported two print-ready STL files and verified manifold mesh edges.
- Prepared print settings, assembly notes, and a physical validation plan.

WHY IT MATTERS

The project connects open hardware with data science around a problem students can investigate locally: the quality of the air in learning spaces. A future electronics prototype could log PM, temperature, and humidity data, visualize trends, and support hands-on STEAM workshops. It is designed as a practical bridge between fabrication, sensing, and evidence-based community action.

HAWAGUARD MINI

DESIGN + BUILD PLAN

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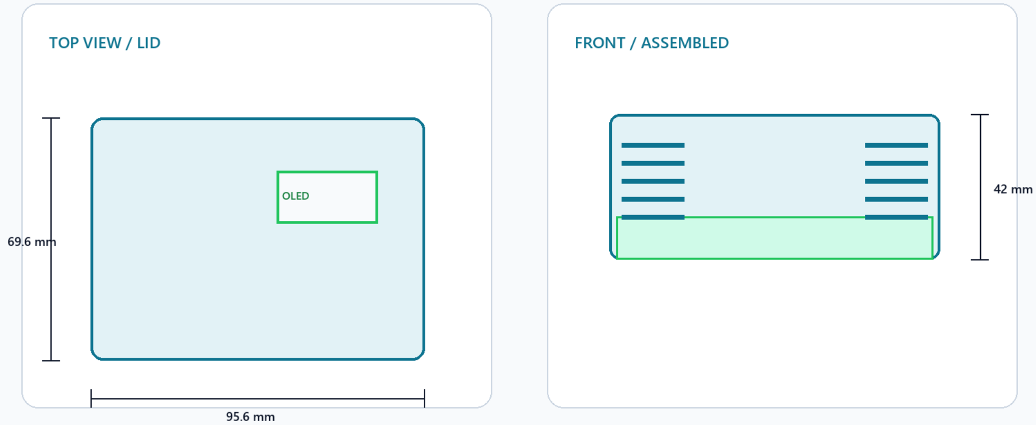
HAWAGUARD MINI - DIMENSIONED DESIGN

Two-part, support-free enclosure for a low-cost ESP32 air-quality monitor



2 PARTS Base tray + vented lid	**2.4 MM** Wall thickness	**0.35 MM** Friction-rib engagement	**NO SUPPORTS** Both parts print flat

95.6 x 69.6 x 42 mm overall envelope	**2.4 mm** nominal walls	**2 parts / no supports** print strategy	**PETG or indoor PLA** prototype material

INTENDED HARDWARE

Controller bay - ESP32 DevKit-style board, up to 29.2 x 51.5 mm
Sensor bay - SPS30-class particulate sensor, up to 42 x 42 mm
Display - 0.96-inch OLED with a 29 x 15 mm window
Additional sensing - compact temperature/humidity module
Power/data - rear 14 mm USB cable relief

RECOMMENDED PRINT SETUP

0.4 mm nozzle | 0.20 mm layers | 3 perimeters | 15-20% gyroid infill | no supports. Print the base with its floor on the build plate; the lid STL is already oriented display-face down.

BUILD AND VALIDATION ROADMAP

1. Requirements + CAD - complete
Defined component bays, vents, OLED opening, cable relief, and tool-free fit.

2. Mesh verification - complete
Both STL surfaces passed an edge-manifold check with zero open/non-manifold edges.

3. Fit coupon + first print - pending
Check lid engagement and adjust XY compensation if required.

4. Electronics + data test - planned
Assemble sensors, compare readings with a reference monitor, then log and visualize classroom trends.

WHAT THIS SAMPLE DEMONSTRATES

Original project framing, AI-assisted functional CAD, fabrication planning, technical honesty, and a data-focused community use case.